

## LULC entities

1. **CHANGE**: Any verb or noun phrase that denotes a measurable difference in land cover extent, quality, or characteristic between two or more temporal observations. CHANGE entities capture both the direction (increase, decrease, loss, gain) and nature (converted, degraded, improvement) of landscape dynamics
2. **LOC**: Any proper noun referring to a geographically identifiable area at any spatial scale, including continents, countries, regions, provinces, cities, villages, or named places where LULC phenomena are observed.
3. **LULC**: Any noun or noun phrase that describes a specific type of land surface characterized by specific biophysical attributes or human utilization patterns. These entities encompass both natural land cover types (e.g., forests, woodlands, savannas) and anthropogenic land uses (e.g., agricultural land, cropped fields, fallows)
4. **DATE**: Any numeric expression or textual reference that specifies a particular year, decade, or other precise temporal point at which a land cover observation was made or a change , process occurred (1990, 2020, January 2000)
5. **PERCENT**: Any numerical expression followed by the percent symbol (%) or textual equivalent (e.g., "half") that represents a proportional measurement relative to a defined whole or baseline.
6. **QUANTITY**: Any numerical value that represents a count, ratio, index, or other non-percentage quantitative measure relevant to land use/land cover analysis.
7. **COORDINATES**: Any numerical expression representing geographic coordinates in a recognized coordinate reference system that specifies an exact location on Earth's surface. (40.7°N, latitude 23.5)
8. **SURFACE\_UNIT**: Any expression that combines a numerical value with a standard unit of area measurement (e.g., km<sup>2</sup>, hectares, acres)
9. **PROCESS**: Any noun or verb phrase that describes a natural phenomenon (e.g., droughts, desertification) and anthropogenic activities (e.g., cultivation, grazing, irrigation), human activity, or socio-ecological interaction that directly or indirectly causes, influences, or results from changes in land use/land cover patterns.
10. **TIME\_PERIODS** : Time spans with beginning and end (1990-2000, between 1995 and 2005, from 2010 to 2020)

**Distinction:** **TIME\_PERIOD** indicates a *duration*, whereas **DATE** pinpoints a *moment* or *specific year*

## LULC relations

| Relationship | Source Entity Type | Target Entity Type | Description | Example (Original Sentence) | Relation |
|--------------|--------------------|--------------------|-------------|-----------------------------|----------|
|--------------|--------------------|--------------------|-------------|-----------------------------|----------|

|                                      |         |                     |                                                                                                                        |                                                                                                                               |                                                              |
|--------------------------------------|---------|---------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>CHANGE_TO</b>                     | LULC    | LULC                | Represents a transformation where one land use/land cover type is converted into another type                          | About 26 and 31% of the total area of natural vegetation (forest, woodland and tree savanna) will be converted to farmland... | forest <b>LULC CHANGE_TO</b> "farmland" <b>LULC</b>          |
| <b>INCREASES/DECREASES</b>           | LULC    | NULL                | Shows the (increase or decrease) in a land cover type, typically expressed as a percentage                             | Forest land was <b>decreased</b> in 2002 compared to its coverage in 1984.                                                    | "Forest land <b>LULC DECREASES NULL</b>                      |
| <b>INCREASES _BY / DECREASES _BY</b> | LULC    | PERCENT or QUANTITY | Shows the magnitude of change (increase or decrease) in a land cover type, typically expressed as a percentage         | Forest land was <b>decreased nearly by</b> half in 2002 compared to its coverage in 1984.                                     | "Forest land <b>LULC DECREASES _BY</b> half" <b>QUANTITY</b> |
| <b>CAUSES</b>                        | PROCESS | CHANGE              | Establishes a causal link between a process and a change. The process is the antecedent, and the change is the result. | After the droughts in the 1970s and 1980s, the observed loss of woody vegetation cover                                        | "droughts <b>PROCESS CAUSES</b> loss" <b>CHANGE</b>          |

|                      |                                           |               |                                                                                                                             |                                                                                                                |                                                                           |
|----------------------|-------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>LOCATED_IN</b>    | LULC, PROCESS, CHANGE                     | LOC           | Specifies the spatial context by indicating the location where a LULC, process, or CHANGE.                                  | The forests of West and Central Africa..."                                                                     | "forests <b>LULC LOCATED_IN</b> West and Central Africa" <b>LOC</b>       |
| <b>OCCURS_DURING</b> | PROCESS, CHANGE,                          | TIME_PERIOD,  | Indicates the temporal context by specifying when a process or change took place. It can link to a specific period of time. | After the droughts in the 1970s and 1980s..                                                                    | "droughts <b>PROCESS OCCURS_DURING</b> 1970s and 1980s <b>TIME_PERIOD</b> |
| <b>ENABLES</b>       | PROCESS                                   | PROCESSES     | One process enables another process                                                                                         | <i>No direct example provided in the text</i>                                                                  | deforestation <b>PROCESS ENABLES</b> urbanization <b>PROCESS</b>          |
| <b>MEASURES</b>      | PERCENT, CARDINAL, QUANTITY, SURFACE_UNIT | CHANGE, LULC. | This is a relation that quantifies, meaning it involves a quantity that serves as a measure or description for some entity. | describes Rwandan agriculture as caught in a 'low investment trap'; less than 10% of cropland is irrigated.... | 10%, <b>PERCENT, MEASURES</b> , cropland, <b>LULC</b>                     |

|                  |                     |         |                                                                                                   |                                                                                                                                                         |                                                                         |
|------------------|---------------------|---------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>AFFECTS</b>   | PROCESS<br>, CHANGE | LULC,   | Describes the impact of a process or change on LULC.                                              | After the droughts in the 1970s and 1980s, the observed loss of woody vegetation cover was often considered as irreversible...                          | "loss <b>change</b> <b>AFFECTS</b> woody vegetation cover <b>LULC</b> " |
| <b>FROM_TO</b>   | PERCENT             | PERCENT | Shows the starting and ending percentage values to describe the range of changes.                 | From 1996 to 2017, the area covered by cropped fields, C, has increased from 40% to 51.3%....                                                           | "from 40% <b>TO</b> 51.3%" - <i>The area changed from 40% to 51.3%.</i> |
| <b>OCCURS_IN</b> | CHANGE,<br>PROCESS  | DATE    | Indicates the temporal context by specifying when a process or change took place in specific date | About 26 and 31% of the total area of natural vegetation (forest, woodland and tree savanna) will be converted to farmland and degraded savanna by 2050 | Converted <b>CHANGE</b> <b>OCCURS_IN</b> 2050 <b>DATE</b>               |